

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (IT) Examination

May 2013

IT703 Web Service Service Oriented Architecture

Date: 13.05.2013, Monday

Time: 10:00 a.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) What is Service oriented architecture? How it is differentiate from 3-tier client server architecture? [03]
- (b) Design a XML file for placing online order for purchasing a book. Check shipping cost. It should not be exceed more than 30Rs. Design XSLT for the same. [04]
- Q - 2 (a) Web service must be highly cohesive and loosely coupled. Justify [02]
- (b) Do as directed: [08]
1. Explain the hierarchy of web services.
 2. What is the role of WSDL in SOA?
 3. Narrate the difference between SOAP based and RESTful web service.
 4. What is the usage of Envelope element in SOAP message format?
- (c) Compare Oracle SOA with IBM SOA. [04]

OR

- Q - 2 (a) Explain the black box nature of web service. [05]
- (b) Write a program to make a login application using AJAX and JavaScript with proper validation. There will be two types of login (Moderator and user login) with different access rights. [07]
- (c) What is the difference between stub and proxy? [02]
- Q - 3 (a) Design RSS feed for making a IPL time table [03]
- (b) What is semantic web? Explain the concept of RDF in detail. [04]
- (c) Explain the SOA platform and describe the API used in J2EE to build SOA. [07]

OR

- Q - 3 (a) How service oriented architecture achieved in .NET? Discuss in detail. [07]
- (b) Define the various characteristics of web services. [02]
- (c) Define the rules for creating any xml file. What is the importance of XML schema? [05]

SECTION – II

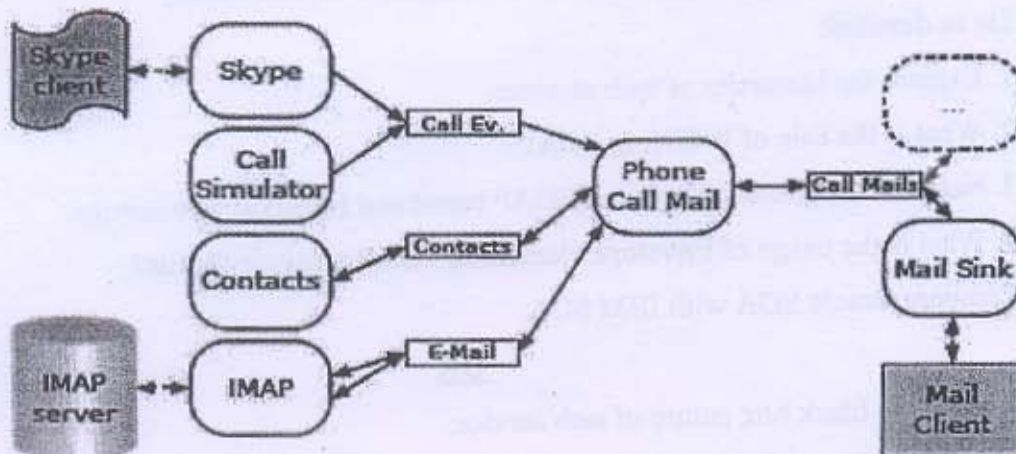
- Q - 4 (a) Draw the complete SOA architecture and compare it with an Enterprise Architecture. [05]
(b) State the difference between fine grain web service and coarse grain web service. [02]
- Q - 5 (a) Develop RMI program for performing a addition of three numbers [07]
(b) Define Orchestration and Choreography in detail with three real life applications. [07]

OR

- Q - 5 (a) What is the role of JSTL in SOA? Write any sample program of JSTL which demonstrate the use of xml library. [07]
(b) What are the benefits of using SOA in any organization? [07]
- Q - 6 (a) What is RMI? Explain architecture of RMI. State at least five application in real life [07]
(b) What is the difference between business logic and presentation logic? Explain in terms of servlet and JSP in detail with example. [07]

OR

- Q - 6 Case Study: Skype Mail [14]



Display email exchange with caller.

Facility provided by above system:

1. heterogeneous data
2. push/pull
3. event triggered
4. processing
5. reusability

Design the web services for the above system and explain how SOA fit in that system.
